Relations and Functions

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March 2025

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1 Important Terms

1. aRb: a is related to b.
2. $\forall$: for all.
3. $\in$: belongs to.
4. $\exists$: there exists.
5. $\Rightarrow$: implies.
6. $\Leftrightarrow$: if and only if.

2 Cartesian Product and Relations

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

A relation from A to B is a subset $R \subseteq A \times B$.

$$R = \{(x, y) : xRy\}.$$

For $A = \{1, 2, 3, 4\}$, $A \times A$ contains the 16 ordered pairs $(1, 1)$ through $(4, 4)$. **Counting note:** If $|A| = n$ and $|B| = m$, then the number of relations from A to B is 2^{nm}, since each subset of $A \times B$ is a relation.

3 Trivial Relations

3.1 Empty Relation

$$R = \varnothing.$$

3.2 Universal Relation

$$R = A \times A.$$

4 Types of Relations

4.1 Reflexive

$$\forall a \in A, \quad (a, a) \in R.$$

For $A = \{1, 2, 3, 4\}$, $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ is reflexive.

4.2 Symmetric

$$(a, b) \in R \Longrightarrow (b, a) \in R.$$

4.3 Transitive

$$(a, b) \in R \text{ and } (b, c) \in R \Longrightarrow (a, c) \in R.$$

4.4 Equivalence Relation

A relation is an equivalence relation if it is reflexive, symmetric and transitive. The equivalence class of x is

$$[x] = \{a \in A : (x, a) \in R\}.$$

5 Types of Functions

5.1 Identity Function

$$f(x) = x.$$

5.2 Constant Function

$$f(x) = c.$$

5.3 Polynomial Function

$$f(x) = a_n x^n + \cdots + a_1 x + a_0.$$

5.4 Rational Function

$$f(x) = \frac{p(x)}{q(x)}, \qquad q(x) \neq 0.$$

For example, $f(x) = 1/x$.

5.5 Modulus Function

$$f(x) = |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

5.6 Signum Function

$$\operatorname{sgn}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases}$$

5.7 Greatest Integer Function

$$f(x) = \lfloor x \rfloor, \qquad \lfloor x \rfloor = n \text{ for } n \leq x < n + 1.$$

Examples: $\lfloor 2.5 \rfloor = 2$, $\lfloor -0.5 \rfloor = -1$.

6 Injective and Surjective Functions

6.1 One to One Function

A function is injective when

$$f(x_1) = f(x_2) \Longrightarrow x_1 = x_2.$$

The horizontal line test checks whether a graph is one to one.

6.2 Onto Function

A function $f : A \to B$ is onto if for every $y \in B$ there exists $x \in A$ with $f(x) = y$. Its range equals its codomain. A function that is both one to one and onto is bijective.

7 Counting Relations and Functions

For a set A of n elements,

$$\text{Relations on } A = 2^{n^2}, \quad \text{Reflexive relations} = 2^{n^2-n}, \quad \text{Symmetric relations} = 2^{n(n+1)/2}.$$

For $|A| = m$, $|B| = n$,

$$\text{Functions } A \to B = n^m, \qquad \text{One to one functions} = \frac{n!}{(n-m)!} \quad (m \leq n).$$